FINM3013 Introduction to Financial Derivatives

Chapter 2 Exercises

Solutions

1.

Open interest is the number of contract outstanding. Many traders close out their positions just before the delivery month is reached. This is why the open interest declines during the month preceding the delivery month. The open interest went down by 600. We can see this in two ways. First, 1,400 shorts closed out and there were 800 new shorts. Second, 1,200 longs closed out and there were 600 new longs.

2.

There is a margin call if $1000 is lost on the contract. This will happen if the price of wheat futures rises by 20 cents from 250 cents to 270 cents per bushel. $1500 can be withdrawn if the futures price falls by 30 cents to 220 cents per bushel.

3.

You could go long one June oil contract and short one December contract. In June you take delivery of the oil borrowing $80 per barrel at 5% to meet cash outflows. The interest accumulated in six months is about $80\times0.05\times0.5$ or $2. In December the oil is sold for $86 per barrel and $82 is repaid on the loan. The strategy therefore leads to a profit of $4. Note that this profit is independent of the actual price of oil in June 2010 or December 2009. It will be slightly affected by the daily settlement procedures.